

Run	Train data (#sample)	CG R@1	WIT R@1	Bloom R@1	Mammoth R@1	ImageNet	R@1-Avg
SEA-CLIP-Tiny	CC12M + CG-OE-filt + WIT-hf + Bloom + Mammoth-VL-SEA (12.72M)	29.3	21.8	23.3	15.3	36.4	17.4
<i>Dataset Ablation</i>							
Only SEA datasets	CG-OE-filt + WIT-hf-base + Bloom (1.21M)	63.7	54.6	24.9	0.4	5.6	9.4
Only CC12M	CC12M (10.97M)	0.1	1.9	1.2	0.5	40.0	6.5
Only WIT	WIT-hf-base (487K)	0.2	27.7	2.2	0.1	1.3	6.6
Only Bloom	Bloom-only (21K)	0.0	0.0	1.0	0.0	0.1	4.7
Only CG	CG-only (703K)	46.5	0.2	0.6	0.0	0.2	5.1
Only Mammoth	SEA-Mammoth (540K)	0.0	0.0	0.8	—	0.5	5.2
CC12M + SEA	CC12M + CG-OE-filt + WIT-hf + Bloom (12.18M)	33.3	25.2	16.1	0.8	37.4	14.2
<i>KD Loss Ablation</i>							
Only CKD	CC12M + CG-OE-filt + WIT-hf + Bloom + Mammoth-VL-SEA (12.72M)	28.5	20.3	11.3	—	32.7	13.7
Only FD	CC12M + CG-OE-filt + WIT-hf + Bloom + Mammoth-VL-SEA (12.72M)	39.5	26.8	11.3	—	31.3	12.1
Only ICL	CC12M + CG-OE-filt + WIT-hf + Bloom + Mammoth-VL-SEA (12.72M)	39.2	26.4	15.3	—	34.2	15.7
w/o KD	CC12M + CG-OE-filt + WIT-hf + Bloom + Mammoth-VL-SEA (12.72M)	39.1	26.1	15.6	—	31.4	12.1
SEA-CLIP-Tiny w/o KD losses	CC12M + CG-OE-filt + WIT-hf + ML (12.33M)	47.4	41.5	28.8	—	—	—
SEA-CLIP-Tiny (CKD only)	CC12M + CG-OE-filt + WIT-hf + Bloom + Mammoth-VL-SEA (12.87M)	35.6	28.0	23.2	—	—	—
Teacher	—	12.9	32.2	9.6	16.6	71.1	46.4

Table 1: The dataset ablation study. All models are trained on the same training pipeline and hyperparameters. SEA-CLIP-Tiny is our full data blend (CC12M + SEA + Mammoth-VL-SEA). The dataset ablation swaps in SEA blend, CC12M, a single SEA source (WIT, Bloom, CG, or Mammoth), or CC12M+SEA as the only training data. We use **R@1-Avg** from Total Average@1 in Table ??.